

Francesco Cavina

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Data and AI engineer with a BSc in Econometrics and Data Science from Vrije Universiteit Amsterdam and incoming MSc studies in Econometrics and Operations Research and Business Analytics. Experience building LLM workflows, RAG prototypes, governed data interfaces, Python data pipelines, APIs and full stack tools. Strong fit for student AI agent engineering work that turns practical workflow problems into agents, integrations, automations and usable internal tools.

TECHNICAL SKILLS

- **AI and LLM Engineering:** LLM workflows, RAG, vector search, ChromaDB, OpenAI API, Hugging Face, sentence transformers, prompt design, structured prompts, local LLM workflows, citation based responses
- **Backend and Data:** Python, FastAPI, SQL, PySpark, REST APIs, SQLAlchemy, data ingestion, schema inference, metadata driven workflows, data quality workflows
- **Frontend and Product:** TypeScript, JavaScript, Next.js, React, Tailwind CSS, HTML, CSS, UI components, user focused prototypes, self service tools
- **Infrastructure:** Git, Docker, Kubernetes, Terraform, Azure, Databricks, Airflow
- **Quantitative Work:** econometrics, probability, statistics, forecasting, model evaluation, walk forward validation, optimisation, machine learning with scikit learn, Random Forest, XGBoost and neural networks

RELEVANT EXPERIENCE

Data & AI Platform Engineer

Robodata
Netherlands

September 2025 - June 2026, Amsterdam,

- Built privacy aware AI and data platform components for enterprise clients, including metadata driven ingestion, schema inference and PII classification workflows.
- Developed automated data pipelines in Python, PySpark and SQL using Bronze, Silver and Gold architecture.
- Built local LLM workflows for data discovery, field classification and metadata generation, with attention to governed use of enterprise data.
- Created AI ready data interfaces that allowed internal datasets to be queried more safely and consistently by LLM based tools and agents.
- Worked on self service data tooling so non specialist users could understand, operate and reuse technical data infrastructure.
- Documented platform components, workflows and design choices to support handover, adoption and future development.
- Used Docker, Kubernetes, Terraform, Azure, Databricks, Airflow and SQLAlchemy across production platform work.

Operational Trader

QuantFi
Netherlands

October 2022 - May 2023, Amsterdam,

- Developed and tested Python based forecasting and trading strategies using Random Forest, XGBoost, neural networks and walk forward validation.
- Built data ingestion, retraining and daily forecast workflows in Python and SQL to support recurring trading decisions.
- Created dashboards and visualisations to compare model signals, realised performance and forecast quality.
- Evaluated model output with attention to overfitting, changing market conditions, assumptions and practical usability.
- Worked in an environment where clarity, disciplined analysis and timely decision making mattered daily.

Teaching Assistant

Vrije Universiteit Amsterdam
Netherlands

June 2022 - November 2024, Amsterdam,

- Taught weekly Probability Theory tutorials to first year Econometrics students.
- Explained abstract mathematical concepts in a clear and practical way, with focus on reasoning rather than memorising procedures.
- Helped students understand the assumptions behind probability exercises and build confidence with formal problem solving.
- Built a student portfolio system with practical tools and information for first year students and represented the EOR programme during Open Days.

EDUCATION

MSc in Econometrics and Operations Research

Track Business Engineering and Data Driven Decision Making

• Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2026-2029

MSc in Business Analytics

Track Computational Intelligence • Vrije Universiteit Amsterdam

• Amsterdam, Netherlands • 2026-2029

Bachelor of Science in Econometrics and Data Science

Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2021-2024 • 7.8/10 •
Thesis: Nonstandard Probabilities in an Internal Probability Space, grade 8.5/10

Minor in Pure Mathematics

University of Leeds • Leeds, England
• 2024 • 8.5/10

PROJECTS

- 1) [AI Powered Personal Website](https://github.com/FrancescoCavina02/My-Website) • <https://github.com/FrancescoCavina02/My-Website> • November 2025 - present
Built a full stack portfolio with a FastAPI backend that parses an Obsidian vault into a REST API for notes, quotes and chat functionality. Used Next.js, TypeScript, Python and Tailwind CSS.
- 2) [RAG Knowledge Chatbot](https://github.com/FrancescoCavina02/Spiritual-chatbot) • <https://github.com/FrancescoCavina02/Spiritual-chatbot> • November 2025 - present
Built a retrieval augmented chatbot using semantic chunking, vector search, structured prompts and citation based responses. Focused on reliable retrieval, traceable answers, hallucination reduction and reusable workflow design. Designed the system around the problem of making a knowledge base queryable by an LLM while preserving source grounding.
- 3) [Dodgeball Club Amsterdam](https://github.com/FrancescoCavina02/DCA-website) • <https://github.com/FrancescoCavina02/DCA-website> • June 2025 - present
Built and maintain a public website using HTML, CSS and JavaScript. Focused on clear navigation, simple content management and practical usability for club members.

Podcasts

[Podcast Link](#) - Created two independent interview podcasts with researchers in physics, cosmology, mathematics and philosophy.

RELEVANT FIT FOR GAIDE

- Practical experience building LLM and data platform components, not only experimenting with AI tools.
- Comfortable working across AI engineering, backend APIs, data workflows, lightweight frontend work and documentation.
- Experience translating unclear workflows into usable prototypes and self service tools.
- Strong quantitative foundation for evaluating models, assumptions and practical output quality.
- Able to explain technical ideas clearly to students and non specialist users.